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VLG-RMOT: Calibration-before-control for end-to-end referring multi-object tracking

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ABSTRACT

Referring multi-object tracking (RMOT) aims to detect and track all objects that satisfy a natural-language expression in video. End-to-end RMOT relies on object queries for detection, association, and temporal updating, so language must function as both an alignment cue and a control signal for evolving queries. Existing methods have advanced visual-linguistic fusion and query interaction; however, alignment alone does not ensure a scene-discriminative language signal as candidate objects, their appearance, and their relations change across frames. We therefore propose VLG-RMOT, a calibration-before-control framework that adapts language to the current visual scene before using it for query control. Specifically, cross-modal semantic calibration performs bidirectional visual-linguistic calibration with learnable channel-wise residual scaling. The resulting scene-conditioned language guides two complementary stages: contextual query synthesis establishes a shared referring prior before visual decoding, while text-aware gating modulates query states after visual cross-attention. On Refer-KITTI, Refer-KITTI-V2, and Refer-BDD, VLG-RMOT achieves the best HOTA among the compared end-to-end methods. Under a matched ablation protocol, learnable calibration improves HOTA by 1.97, 1.40, and 2.98 points over raw-language control, an unscaled residual, and fixed scaling, respectively; scaling-vector statistics and token-level visualization provide representation-level evidence of channel-selective and scene-dependent adaptation.

1. Introduction

Referring multi-object tracking (RMOT) represents a fundamental challenge in modern computer vision, aiming to detect and track multiple objects in video sequences based on natural language descriptions (Wu et al., 2023). This task is particularly important for developing artificial intelligence systems that can interact with the visual world through natural language, providing greater flexibility and control compared to traditional multi-object tracking methods. Despite being an emerging research area, RMOT has demonstrated extensive potential for applications across diverse scenarios, including autonomous driving (Lin et al., 2024; Lu et al., 2025), multimodal agent (Noh et al., 2025), and human-object interaction (Li et al., 2025b).

End-to-end RMOT integrates language grounding, object detection, and temporal association within a unified tracking-by-attention pipeline (Chen et al., 2025; He et al., 2024; Liang et al., 2025; Ma et al., 2024; Wu et al., 2023; Xiao et al., 2026; Zhang et al., 2025). As illus-

trated in Fig. 1(a), a generic framework encodes the referring expression and video frames, fuses their representations in a multimodal encoder, and refines generic object queries through a standard decoder to predict referred tracks. This generic formulation provides the common architectural basis considered in this work, but it does not explicitly expose scene-calibrated language as a control signal for query initialization and post-visual refinement. Fig. 1(b) shows the architecture of VLG-RMOT: CSC first produces scene-conditioned visual and linguistic features, CQS uses calibrated language for query-context control, and TAG applies post-visual control within the decoder. Together, these components establish a calibration-before-control pathway from scene-conditioned representation learning to query initialization and post-visual decoder refinement.

However, a fundamental issue remains in end-to-end RMOT: cross-modal alignment does not automatically make language a reliable control signal for object queries. Existing methods have substantially advanced visual-linguistic fusion, query interaction, semantic reasoning,

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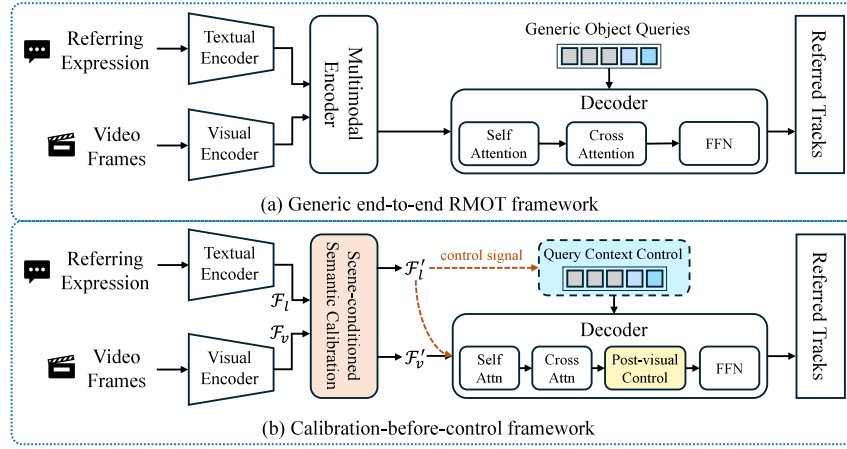


Fig. 1. Architectural comparison between a generic end-to-end RMOT framework and VLG-RMOT. (a) A generic end-to-end framework fuses textual and visual features in a multimodal encoder and refines generic object queries with a standard decoder. (b) VLG-RMOT introduces scene-conditioned semantic calibration, calibrated query-context control, and post-visual control, corresponding to CSC, CQS, and TAG, respectively.

and temporal modeling (He et al., 2024; Liang et al., 2025; Ma et al., 2024; Wu et al., 2023; Xiao et al., 2026; Xu & Huang, 2026; Zhang et al., 2025). These advances improve correspondence between modalities, but RMOT imposes an additional requirement on the linguistic representation used by the tracker. It must remain discriminative as the number, appearance, and motion of candidate objects change across frames, because language-guided query states support both the discovery of referred targets and the maintenance of identity-consistent trajectories. Consequently, the importance of each word is inherently scene-dependent (Ding et al., 2023). For example, “black” is useful when color distinguishes the visible candidates, while “left” must be interpreted relative to the current frame layout. If the semantic emphasis of language does not adapt to the current visual competition, visually salient but expression-inconsistent evidence may dominate query updates, causing false positives, missed targets, or identity ambiguity. This leads to the central question: **how can language be calibrated against the current visual evidence before it is used to control query-based tracking?**

To address this issue, we propose VLG-RMOT, a **calibration-before-control** framework that turns visually calibrated language into an explicit control signal for end-to-end RMOT. The central idea is to establish a reliable linguistic representation before it participates in query initialization and iterative decoder updating. Cross-modal semantic calibration (CSC) performs bidirectional visual-linguistic calibration, with learnable scaling factors regulating the residual information exchange so that language can absorb scene-specific evidence while retaining its original semantics. Contextual query synthesis (CQS) converts the calibrated representation into a compact global semantic prior and injects it into learnable content queries before visual decoding. Text-aware gating (TAG) then reuses the calibrated language after visual cross-attention to suppress query responses that are inconsistent with the referring expression. Together, CSC, CQS, and TAG establish a coherent calibration-before-control pathway that links scene-aware language calibration to query initialization and post-visual refinement.

Our main contributions are summarized as follows:

- We present a calibration-before-control design for end-to-end RMOT, where language representations are first adapted to the current visual scene before being used to guide query-based tracking.
- We instantiate this design with CSC, CQS, and TAG, which respectively perform scene-conditioned language calibration, calibrated semantic query initialization, and post-visual language-aware query modulation.
- We provide empirical evidence through benchmark comparisons, ablation studies, and visualizations that vision-calibrated language is

more effective than static language cues for guiding object queries in end-to-end RMOT.

2. Related work

2.1. Referring multi-object tracking

Referring multi-object tracking (RMOT) aims to detect and track all objects that satisfy a natural-language expression in a video (Wu et al., 2023). Unlike conventional multi-object tracking (Meinhardt et al., 2022; Xie et al., 2024a,b; Zhang et al., 2022), RMOT uses language rather than predefined categories to specify the targets of interest (Li et al., 2025d). The development of RMOT can be organized according to when language enters the tracking pipeline and which component it is intended to improve.

Trajectory-first matching. Two-stage methods first obtain object trajectories with a unimodal tracker and then retrieve those that match the expression (Du et al., 2024; Li et al., 2025c,d). iKUN (Du et al., 2024) introduces an insertable knowledge-unification module and a neural Kalman filter, while CPAny (Li et al., 2025c) improves compatibility with different visual and textual encoders. This line can reuse mature tracking systems and flexible encoders, but language is introduced only after visually generated trajectories are available. Its recall is therefore bounded by the quality and coverage of proposals produced without the referring constraint.

Unified visual-language fusion. TransRMOT (Wu et al., 2023) establishes the end-to-end tracking-by-attention paradigm for RMOT and uses early fusion to incorporate linguistic cues into visual features. DeepRMOT (He et al., 2024), MLS-Track (Ma et al., 2024), and HFF-Tracker (Zhao et al., 2025) subsequently strengthen cross-modal representation learning through deep, multi-level, or hierarchical fusion. These methods jointly optimize grounding and tracking and avoid post-hoc trajectory retrieval. Their principal emphasis, however, is on enriching fused representations and decoder interaction, rather than explicitly modeling the reliability of scene-conditioned language as a control signal that is reused before and after visual query interaction.

Query-centric and temporal enhancement. A second end-to-end line improves how object queries are initialized, propagated, or associated. MGLT (Chen et al., 2025) uses multi-granularity query bootstrapping and track-prompt alignment, while collaborative matching (Huang et al., 2026) addresses the imbalance between newborn and existing targets. TempRMOT (Zhang et al., 2025) introduces query-driven temporal enhancement to aggregate historical information, and LTTrack (Xu & Huang, 2026) combines explicit language guidance with short-

term and long-term temporal representations. These approaches improve query persistence and temporal robustness. Their main advances lie in query construction or memory-based temporal modeling, which is complementary to calibrating the language signal against the current set of visual candidates before it controls query updates.

Structured semantic guidance. Recent methods also introduce explicit structure into the referring expression. DKGTrack (Li et al., 2025a) separates localized descriptions from motion states for stage-specific guidance. CDRMT (Liang et al., 2025) decomposes language into static object attributes and spatial-motion information and progressively injects these cues into object queries. Such decomposition provides interpretable, fine-grained guidance, but it addresses a different axis from VLG-RMOT and relies on an explicit semantic factorization of the expression. VLG-RMOT instead calibrates the complete language representation with the observed scene and uses the resulting features to control query initialization and post-visual refinement.

Positioning of VLG-RMOT. VLG-RMOT follows the end-to-end tracking-by-attention formulation introduced by TransRMOT (Wu et al., 2023), but changes the role of language from an early fusion cue to a scene-conditioned query-control signal. TempRMOT (Zhang et al., 2025) and LTTrack (Xu & Huang, 2026) primarily strengthen temporal query representations and memory, whereas VLG-RMOT targets the reliability of language guidance under the visual competition in the current frame. CDRMT (Liang et al., 2025) improves query learning through explicit semantic decomposition, while VLG-RMOT connects visually calibrated language to two complementary query stages through CQS and TAG. These directions are complementary: the specific contribution of VLG-RMOT is a calibration-before-control pathway that links scene-aware language adaptation to both query initialization and post-visual decoder refinement.

2.2. Vision-language alignment and semantic fusion

General alignment and fusion. Vision-language learning has progressed from two-tower matching to deep multimodal interaction. Early methods process visual and textual inputs separately and perform shallow matching at a later stage (Liu et al., 2019a; Sadhu et al., 2020). Transformer-based fusion encoders subsequently enable token-level interaction through self-attention or cross-modal attention (Dai et al., 2025, 2024; Song et al., 2021; Vaswani et al., 2017), including early-, late-, mutual-, and joint-attention designs (Guo et al., 2022; Liu et al., 2019b; Shao et al., 2024; Zhou et al., 2023). Vision-language pre-training models such as BEiT-3 (Wang et al., 2023), CLIP (Radford et al., 2021), and ALBEF (Li et al., 2021) further provide transferable multimodal representations for downstream tasks (Li et al., 2025c). This progression has substantially improved cross-modal correspondence, but does not by itself determine how the aligned language representation should control the evolving queries of an online tracker.

Language-conditioned video reasoning. Referring video segmentation provides related mechanisms for temporal language guidance. ReferFormer (Wu et al., 2022) formulates language as object queries, language-aware spatiotemporal collaboration updates spatially and temporally relevant language features (Hui et al., 2023), and Locater (Liang et al., 2023) uses local-global memory for efficient video querying. These methods demonstrate the value of language-conditioned video reasoning, but they mainly predict masks for referred objects. RMOT additionally requires online multi-object discovery, identity association, and persistent query updating as the set of referred targets changes over time.

From alignment to query control. The above studies show how to align modalities or inject language into video models. VLG-RMOT focuses on the next question specific to query-based RMOT: how to make the language representation reliable before it controls target queries, and how to preserve that referring constraint after queries interact with visual evidence. CSC, CQS, and TAG address these two requirements

as a coordinated pathway, converting scene-conditioned language into explicit control for query initialization and post-visual refinement.

3. The proposed method

3.1. Preliminary

Problem Formulation. Given a video sequence $\mathcal{V} = \{I_1, I_2, \dots, I_T\}$ consisting of T image frames and a textual query Q describing objects of interest, referring multi-object tracking aims to generate a set of trajectories $\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_K\}$ for all objects matching the description across the video sequence. Each trajectory τ_i is represented as:

$$\tau_i = \{(b_{t_1}^i, o_{t_1}^i, r_{t_1}^i), (b_{t_2}^i, o_{t_2}^i, r_{t_2}^i), \dots, (b_{t_m}^i, o_{t_m}^i, r_{t_m}^i)\}, \quad (1)$$

where $b_t^i = (x_t^i, y_t^i, w_t^i, h_t^i)$ denotes the bounding box coordinates of object i in frame t , o_t^i represents the objectness score indicating the confidence of object presence, r_t^i is the referring score measuring how well the object aligns with the textual query, and $(t_1, t_2, \dots, t_m) \subseteq \{1, 2, \dots, T\}$ is the subset of frames where the object appears.

We formulate RMOT as an end-to-end tracking-by-attention problem that processes frames sequentially in an online manner. The mapping function Φ is defined as:

$$\mathcal{T} = \Phi(I_1, I_2, \dots, I_T; Q). \quad (2)$$

The trajectories are generated through an autoregressive track query mechanism that links corresponding objects across frames. At the core of this approach, the model dynamically maintains and updates a set of track queries $\mathcal{Q}_{\text{track}} = \{q_1, q_2, \dots, q_K\}$ that evolve over time. By leveraging attention mechanisms, the model unifies detection, association, and language grounding within a single architecture, thereby eliminating the need for explicit multi-stage pipelines. This unified framework ensures identity consistency while adapting to dynamic object counts across temporal sequences.

Model Overview. As shown in Fig. 2, VLG-RMOT builds on the standard end-to-end tracking-by-attention architecture and introduces a coordinated language-guidance pathway around its query decoder. At each frame, CSC performs bidirectional calibration between visual and linguistic features and produces a scene-conditioned language representation. This shared representation drives two complementary modules. Before visual decoding, CQS pools the calibrated language into a global semantic prior and injects it into learnable content queries, giving them a referring-aware starting state without replacing the query embeddings. During decoder refinement, TAG uses the same calibrated language after visual cross-attention to filter expression-inconsistent responses from visually updated queries. Together, these stages establish a continuous calibration-before-control pathway in which scene-conditioned language guides both the initialization and post-visual refinement of tracking queries.

3.2. Cross-modal semantic calibration

At each time step t , a pretrained language model is employed to encode the textual query Q with L words into a language embedding $F_l \in \mathbb{R}^{L \times D}$, where D denotes the dimensionality of the embedding vector. Simultaneously, a CNN-based backbone is used to extract the visual pyramid features from the current frame I_t , represented as $f_v^l \in \mathbb{R}^{H^l \times W^l \times C}$, where H^l and W^l refer to the height and width of the l^{th} level feature map, respectively. Then, the visual embedding $F_v \in \mathbb{R}^{N_v \times C}$ is obtained by flattening and concatenating the feature maps across different levels, where N_v corresponds to the total number of visual tokens.

The purpose of CSC is to make the language representation responsive to the current visual scene while retaining an explicit residual path for the original sentence features. Features extracted by a pretrained language encoder are independent of the current frame, whereas directly adding a full cross-modal response may introduce background content

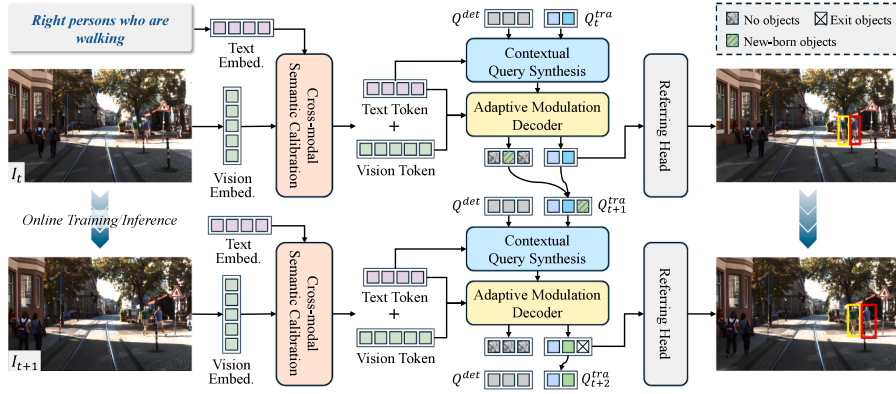


Fig. 2. Overview of VLG-RMOT. The framework follows a calibration-before-control design: cross-modal semantic calibration first adapts language representations to the visual scene, contextual query synthesis uses the calibrated language to initialize learnable content queries, and the adaptive modulation decoder applies text-aware gating after visual interaction to refine query updates.

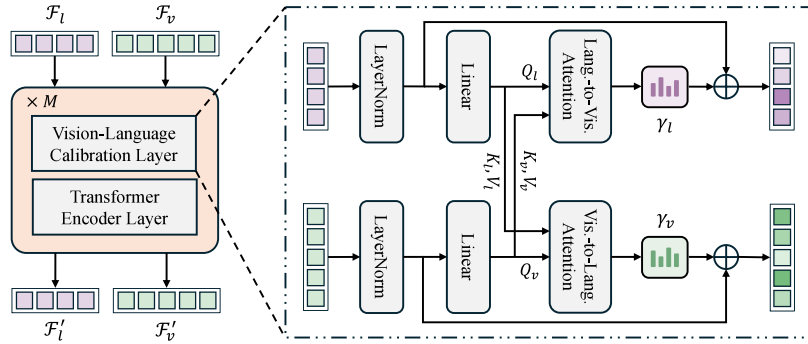


Fig. 3. Illustration of cross-modal semantic calibration. Bidirectional cross-modal attention estimates residual corrections between the two modalities, while the learnable channel-wise vectors γ_v and γ_l regulate how strongly these corrections modify the visual and linguistic streams. The calibrated language features are then used as scene-conditioned control signals for subsequent query initialization and refinement.

or transient distractors. CSC addresses this tension by computing bidirectional cross-modal corrections and injecting them through learnable residual scaling factors. The factors γ_v and γ_l are integral to the calibration process: cross-modal attention determines what information is exchanged, while the scaling factors determine how strongly these corrections modify the visual and linguistic streams. In particular, γ_l controls the contribution of visual evidence to the language representation, allowing scene-supported distinctions to be emphasized while the identity path retains the original language features. Likewise, γ_v regulates the complementary linguistic correction to visual features. Learned scaling therefore allows the model to determine an appropriate calibration strength from data instead of imposing an unscaled or fixed-strength update. The resulting language representation combines scene-sensitive corrections with the original sentence features, making it a reliable control signal for the subsequent CQS and TAG modules.

To this end, we introduce a Cross-modal Semantic Calibration (CSC) module, as depicted in Fig. 3. CSC is placed before query decoding so that the language features used by later query-control modules are first adapted to the observed visual content. The module is constructed by stacking visual-language calibration layers with Transformer encoder layers. Leveraging cross-modal attention, CSC facilitates semantic alignment between visual and linguistic features and produces scene-conditioned language representations for subsequent query initialization and refinement. Concretely, the visual features F_v and linguistic features F_l are first linearly projected into a shared embedding space of dimension d , yielding query, key, and value vectors: $Q_v/K_v/V_v$ for the visual modality and $Q_l/K_l/V_l$ for the language modality. Subsequently, two cross-modal responses are computed. $A_{v \rightarrow l}$ uses visual queries to at-

tend to language keys and values, whereas $A_{l \rightarrow v}$ uses language queries to attend to visual keys and values. The first response remains aligned with the visual tokens and provides their language-conditioned correction, while the second remains aligned with the linguistic tokens and provides their visual-conditioned correction.

$$A_{v \rightarrow l} = \text{softmax}\left(\frac{Q_v K_l^T}{\sqrt{d}}\right) V_l, \quad (3)$$

$$A_{l \rightarrow v} = \text{softmax}\left(\frac{Q_l K_v^T}{\sqrt{d}}\right) V_v.$$

The two responses are mapped back to the channel dimensions of their corresponding residual streams by the attention output projections, which are omitted from the notation for brevity. Finally, the aligned visual features $\hat{F}_v$ and language features $\hat{F}_l$ are derived through residual connections, formulated as:

$$\begin{aligned} \hat{F}_v &= F_v + \gamma_v \odot A_{v \rightarrow l}, \\ \hat{F}_l &= F_l + \gamma_l \odot A_{l \rightarrow v}, \end{aligned} \quad (4)$$

where $\gamma_v \in \mathbb{R}^C$ and $\gamma_l \in \mathbb{R}^D$ are learnable channel-wise scaling vectors, and $\odot$ denotes element-wise multiplication with broadcasting over the token dimension. In our implementation, $C = 256$ and $D = 768$.

The proposed calibration enables mutual enhancement between visual and linguistic features. Visual features are re-weighted based on linguistic cues to emphasize regions related to the expression, while linguistic features are refined by incorporating visual context from candidate objects and scene layout. In the VLG-RMOT pipeline, the latter effect is particularly important: the calibrated language features become

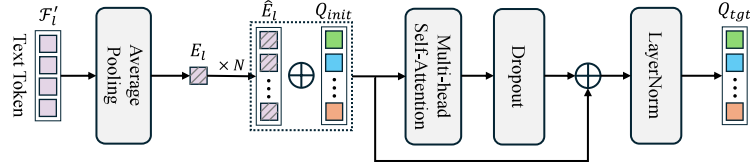


Fig. 4. Illustration of contextual query synthesis. CQS injects semantic priors from calibrated language features into learnable content queries, providing a scene-conditioned starting point for subsequent visual decoding without replacing the query embeddings.

the control signal used by CQS and TAG, allowing query guidance to depend on the current visual evidence rather than on raw sentence embeddings alone.

3.3. Contextual query synthesis

In transformer-based tracking frameworks, the decoder uses object queries to probe encoded visual features. Within RMOT, these queries must not only localize objects but also represent the targets specified by the referring expression. Analogous to prior works (Li et al., 2022; Zhang et al., 2023a; Zhu et al., 2021) on DETection TRansformers (DETR), object queries are composed of content queries and positional queries. Content queries capture semantic, appearance, and category-related information, while positional queries encode spatial location and box-related information.

Existing RMOT trackers commonly retain generic DETR-style queries (He et al., 2024; Wu et al., 2023; Zhang et al., 2025), while language-conditioned query construction has been explored in referring models (Ding et al., 2023; Liang et al., 2025; Wu et al., 2022). The relevant question for CQS is therefore not whether language can be injected into queries, but how scene-calibrated language can provide a common referring context without removing the distinct query slots needed to discover multiple matching instances. As shown in Fig. 4, CQS summarizes CSC-calibrated language into a shared semantic prior and adds this prior to the retained learnable content queries before visual cross-attention. This gives every query the same scene-conditioned starting context while preserving the query-specific hypotheses required for multi-object discovery.

CQS realizes the query-initialization branch of calibration-before-control. CSC has already embedded scene-dependent corrections into the language tokens; CQS therefore requires a stable expression-level summary rather than another round of query-dependent token interaction. Average pooling allows the complete set of calibrated tokens to contribute to a single prior E_l without introducing an additional token selector. CQS then adds E_l to the retained content-query bank as a shared initialization signal rather than generating or replacing the queries. Subsequent self-attention and visual cross-attention remain responsible for coordinating the conditioned queries and resolving object-specific hypotheses.

Mathematically, the vision-conditioned language features $\mathcal{F}'_l$ are average-pooled and mapped to the query dimension, yielding a global language embedding $E_l \in \mathbb{R}^C$. This global embedding E_l is then replicated to match the dimensions of the initial content queries $Q_{init} \in \mathbb{R}^{N \times C}$, generating the expanded embedding $\hat{E}_l \in \mathbb{R}^{N \times C}$. The initial fusion process can be subsequently formulated as:

$$Q_{fused}^{(0)} = Q_{init} + \hat{E}_l. \quad (5)$$

Here, N is the number of object queries, and C is the feature dimension. $Q_{fused}^{(0)}$ represents the initial set of language-infused queries.

The shared additive form serves the multi-object nature of RMOT. Since every query receives the same prior, for any pair of queries i and j , $Q_{fused,i}^{(0)} - Q_{fused,j}^{(0)} = Q_{init,i} - Q_{init,j}$; thus, the injection preserves their pairwise differences exactly before self-attention. CQS therefore shifts the query set toward a common referring context without collapsing the distinctions among the retained query slots in Q_{init} .

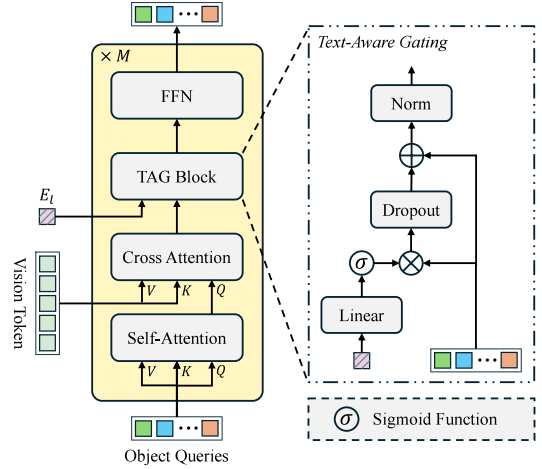


Fig. 5. Illustration of the adaptive modulation decoder. TAG generates gates from calibrated language embeddings and applies them after visual interaction, allowing object queries to preserve expression-relevant responses and suppress language-irrelevant visual evidence during iterative refinement.

To allow the fused representations within each query to interact and refine themselves based on the combined visual and linguistic cues, a multi-head self-attention (MSA) mechanism is applied to $Q_{fused}^{(0)}$. Following standard transformer practices, this is typically followed by dropout, a residual connection, and layer normalization (LN):

$$Q_{tgt} = \text{LN} \left(Q_{fused}^{(0)} + \text{Dropout} \left(\text{MSA} \left(Q_{fused}^{(0)} \right) \right) \right), \quad (6)$$

where $Q_{tgt} \in \mathbb{R}^{N \times C}$ are the final initialized queries, ready to be fed into the subsequent layers of the transformer decoder.

3.4. Adaptive modulation decoder

In transformer-based visual detection frameworks such as DETR (Carion et al., 2020), the decoder serves a critical function in translating global image representations into instance-specific predictions. It takes encoded visual features from the backbone along with a set of learned object queries as input. Through an iterative refinement process involving layers of self-attention for inter-query communication and cross-attention for probing visual features, the decoder progressively transforms these initially generic queries into specialized representations, each focusing on a potential object instance and its visual characteristics, ultimately leading to bounding box and class predictions.

When adapting this architecture for referring multi-object tracking, the decoder must refine object queries under both visual and linguistic constraints. Cross-attention is effective for collecting visual evidence from encoded features, but the most visually salient evidence is not necessarily the one described by the referring expression. After cross-attention, a query may therefore contain both expression-relevant and expression-irrelevant visual responses. This motivates an additional post-visual modulation step that uses calibrated language to regulate query updates after the query has interacted with visual features.

To this end, we propose a text-aware gating (TAG) block, as shown in Fig. 5. TAG operates within the decoder after visual context has been integrated by cross-attention. By leveraging the calibrated language embedding to compute dynamic gates, it modulates the information flow of each query update. This placement follows the calibration-before-control design: the gate is generated from scene-conditioned language, and it is applied after visual interaction so that language can selectively preserve expression-relevant visual responses while attenuating irrelevant ones.

The text-aware gating block operates by first computing a gating signal, denoted as $\mathcal{G} \in \mathbb{R}^{N \times C}$, derived from the global language embedding $E_l \in \mathbb{R}^C$. To achieve dimensional compatibility with object queries, the language embedding is initially broadcast or expanded to $\hat{E}_l \in \mathbb{R}^{N \times C}$. Subsequently, $\hat{E}_l$ undergoes a linear transformation, parameterized by W_g and b_g , followed by a sigmoid activation function σ :

$$\mathcal{G} = \sigma(W_g \hat{E}_l + b_g). \quad (7)$$

The sigmoid function ensures the gating values are between 0 and 1.

Following this gating computation, the TAG block modulates $Q'_{\text{inter}} \in \mathbb{R}^{N \times C}$, the intermediate query representation produced by the decoder cross-attention step:

$$Q''_{\text{inter}} = \text{LN}\left(Q'_{\text{inter}} + \text{Dropout}(\mathcal{G} \odot Q'_{\text{inter}})\right), \quad (8)$$

where LN represents layer normalization, and $\odot$ is element-wise multiplication. The final representation Q''_{inter} proceeds as input to the Feed-Forward Network (FFN) within the decoder.

Conceptually, TAG enables the referring expression to regulate each query after the query has gathered visual evidence. This post-cross-attention placement differs from using language only before decoding: it allows calibrated language to correct the direction of iterative refinement whenever visual responses are strong but not aligned with the expression, leading to more precise localization and attribute recognition.

3.5. Loss function

Following (Wu et al., 2023; Zeng et al., 2022), our framework processes video clips as input during training. The training process involves iterating through frames within a clip, aiming to capture richer temporal information and enhance the capacity to handle challenges such as occlusion and illumination variation. To achieve this, we decompose the predictions and their corresponding losses into two primary components associated with detect queries and track queries. Detect queries are primarily responsible for identifying newly appearing objects in the current frame, while track queries maintain and update representations of targets continuously tracked from previous frames. This decoupling of queries and iterative update mechanism is crucial for achieving end-to-end tracking.

Specifically, for loss calculation at frame t , we differentiate between the processing of detect queries and track queries. For detect queries, we employ bipartite matching to associate their predictions with the ground truth of newly appearing objects in the current frame, a process similar to the object detection loss in DETR (Carion et al., 2020). After matching, the frame-level detection loss $\mathcal{L}_t^{\text{det}}$ is calculated, which includes classification loss $\mathcal{L}_{\text{cls}}$, bounding box L1 loss $\mathcal{L}_{L1}$, generalized IoU loss $\mathcal{L}_{\text{GloU}}$, and referring loss $\mathcal{L}_{\text{ref}}$:

$$\mathcal{L}_t^{\text{det}} = \sum_{i=1}^{N_t^{\text{det}}} \left(\lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{L1} \mathcal{L}_{L1} + \lambda_{\text{GloU}} \mathcal{L}_{\text{GloU}} + \lambda_{\text{ref}} \mathcal{L}_{\text{ref}} \right), \quad (9)$$

where N_t^{det} is the number of newborn objects in frame t , and λ terms are weight coefficients. We set $\lambda_{\text{cls}} = 2$, $\lambda_{L1} = 5$, $\lambda_{\text{GloU}} = 2$, and $\lambda_{\text{ref}} = 2$ in all experiments.

For track queries, we adopt an object consistency matching strategy, directly matching them with their corresponding ground truth targets from the previous frame, eliminating the need to recalculate bipartite

matching with all ground truths. The temporal-level loss $\mathcal{L}_t^{\text{tr}}$ for track queries is calculated similarly to that for detect queries:

$$\mathcal{L}_t^{\text{tr}} = \sum_{i=1}^{N_t^{\text{tr}}} \left(\lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{L1} \mathcal{L}_{L1} + \lambda_{\text{GloU}} \mathcal{L}_{\text{GloU}} + \lambda_{\text{ref}} \mathcal{L}_{\text{ref}} \right), \quad (10)$$

where N_t^{tr} is the number of tracked objects in frame t .

The final training loss is the sum of $\mathcal{L}_t^{\text{det}}$ and $\mathcal{L}_t^{\text{tr}}$ across all frames in the video clip, normalized by the total number of ground-truth objects:

$$\mathcal{L}_{\text{final}} = \frac{\sum_{t=1}^T (\mathcal{L}_t^{\text{det}} + \mathcal{L}_t^{\text{tr}})}{\sum_{t=1}^T (N_t^{\text{det}} + N_t^{\text{tr}})}, \quad (11)$$

where T is the length of a video clip.

4. Experiments

4.1. Datasets

We conduct experiments on three publicly available RMOT datasets. Refer-KITTI (Wu et al., 2023) is the first benchmark dataset for referring multi-object tracking tasks, containing 818 expressions across 18 videos, with 15 videos for training and 3 for testing. The scenes encompass pedestrian streets, public roads, highways, and other diverse environments. Refer-KITTI-V2 (Zhang et al., 2025) addresses the limitations of existing RMOT datasets by introducing more diverse language descriptions and a broader range of visual content. It contains 17 training videos with 8873 expressions and 4 testing videos with 897 expressions, significantly expanding the scale and diversity of the dataset. Refer-BDD (Lin et al., 2024) is designed for autonomous-driving RMOT and contains 50 videos, with 42 used for training and 8 for testing, covering 4610 expressions across diverse urban driving scenes and traffic participants, with more than 100 objects per video on average.

To ensure fair comparison, we adopt Higher Order Tracking Accuracy (Luiten et al., 2021) (HOTA) as the primary evaluation metric, following (Wu et al., 2023). HOTA is a unified tracking metric that balances both detection and association accuracy by integrating detection precision with long-term track consistency in multi-object tracking evaluation.

4.2. Implementation details

VLG-RMOT is implemented using PyTorch. We adopt RoBERTa (Liu et al., 2019c) and ResNet-50 (He et al., 2016) as backbone networks to extract linguistic and visual features, respectively. Deformable DETR (Zhu et al., 2021) is used as the basic framework for detection, and is modified based on our proposed calibration-before-control approach. During training, the weights of the text encoder are frozen, and the rest are initialized using the official weights of Deformable DETR pre-trained on the COCO dataset (Lin et al., 2014). We use a training clip length of $T = 5$ frames. The AdamW optimizer is used to train VLG-RMOT, with a base learning rate of $1e-4$ and a backbone network learning rate of $1e-5$. For Refer-KITTI and Refer-BDD, the benchmark models are trained for 100 epochs, with the learning rate reduced by a factor of 10 at the 50th epoch. The key-component and efficiency analyses on Refer-KITTI use the corresponding 100-epoch checkpoints. For Refer-KITTI-V2, we train for 60 epochs and reduce the learning rate at the 40th epoch, following (Zhang et al., 2025). The focused design-choice ablations on Refer-KITTI use a matched 60-epoch protocol. All other hyperparameters and model settings remain identical to the benchmark experiments. The entire training process is conducted on two 4090 GPUs with a batch size of 1.

4.3. Performance on benchmarks

Table 1 presents the experimental results on the Refer-KITTI (Wu et al., 2023) dataset. VLG-RMOT achieves the best results among end-to-end methods on the main metrics, including HOTA, DetA, and AssA. It

Table 1
Quantitative results on Refer-KITTI.

Method	HOTA	DetA	AssA	DetRe	DetPr	AssRe	AssPr	LocA
Zero-Shot Methods								
Florence2 _{large} (Xiao et al., 2024)	23.46	11.71	47.02	29.54	16.14	51.14	85.30	89.63
Grounding DINO _{base} (Liu et al., 2024)	25.10	11.61	54.42	35.68	14.57	68.25	74.56	87.64
Two-Stage Methods								
FairMOT (Zhang et al., 2021)	22.78	14.43	39.11	16.44	45.48	43.05	71.65	74.77
DeepSORT (Wojke et al., 2017)	25.59	19.76	34.31	26.38	36.93	39.55	61.05	71.34
ByteTrack (Zhang et al., 2022)	24.95	15.50	43.11	18.25	43.48	48.64	70.72	73.90
CStrack (Liang et al., 2022)	27.91	20.65	39.10	33.76	32.61	43.12	71.82	79.51
TransTrack (Sun et al., 2021)	32.77	23.31	45.71	32.33	42.23	49.99	78.74	79.48
TrackFormer (Meinhardt et al., 2022)	33.26	25.44	45.87	35.21	42.10	50.26	78.92	79.63
iKUN (Du et al., 2024)	48.84	35.74	66.80	51.97	52.25	72.95	87.09	-
CPAny (Li et al., 2025c)	53.83	43.40	66.86	60.98	58.69	72.26	88.07	90.85
End-to-End Methods								
TransRMOT ^a (Wu et al., 2023)	46.56	37.97	57.33	49.69	60.10	60.02	89.67	90.33
CO-MOT (Yan et al., 2023)	45.22	33.90	60.45	-	-	-	-	-
EchoTrack (Lin et al., 2024)	39.47	31.19	51.56	42.65	48.86	56.68	81.21	79.93
DeepRMOT (He et al., 2024)	39.55	30.12	53.23	41.91	47.47	58.47	82.16	80.49
MLS-Track (Ma et al., 2024)	49.05	40.03	60.25	59.07	54.18	65.12	88.12	-
TenRMOT (Xiao et al., 2026)	49.77	40.79	60.89	52.65	62.81	65.38	89.28	90.69
MGLT (Chen et al., 2025)	49.25	37.09	65.50	-	-	-	-	-
CDRMT (Liang et al., 2025)	49.35	40.34	60.56	54.54	59.30	64.70	89.80	90.61
LTTrack (Xu & Huang, 2026)	50.91	39.64	65.52	51.99	61.06	69.99	88.59	90.84
VLG-RMOT (Ours)	52.70	40.91	67.95	54.33	60.93	72.88	89.08	91.07

^a Represents the results after frame correction. **Bold** indicates the best result among end-to-end methods.

Table 2
Quantitative results on Refer-KITTI-V2. **Bold** indicates the best result among end-to-end methods.

Method	HOTA	DetA	AssA	DetRe	DetPr	AssRe	AssPr	LocA
Zero-Shot Methods								
Florence2 _{large} (Xiao et al., 2024)	19.03	10.11	35.98	30.03	13.12	39.75	80.78	87.23
Grounding DINO _{base} (Liu et al., 2024)	20.36	9.57	43.41	46.14	10.71	52.11	72.80	87.52
Two-Stage Methods								
FairMOT (Zhang et al., 2021)	22.53	15.80	32.82	20.60	37.03	36.21	71.94	78.28
ByteTrack (Zhang et al., 2022)	24.59	16.78	36.63	22.60	36.18	41.00	69.63	78.00
iKUN (Du et al., 2024)	10.32	2.17	49.77	2.36	19.75	58.48	68.64	74.56
CPAny (Li et al., 2025c)	42.81	31.35	58.61	55.89	40.77	63.78	85.71	89.22
End-to-End Methods								
TransRMOT (Wu et al., 2023)	31.00	19.40	49.68	36.41	28.97	54.59	82.29	89.82
CDRMT (Liang et al., 2025)	31.99	20.37	50.35	26.40	46.26	53.40	85.90	90.36
CGATracker (Zhuang et al., 2025)	33.19	22.04	50.13	38.00	33.88	54.73	84.90	-
LTTrack (Xu & Huang, 2026)	34.64	23.30	51.68	35.17	40.07	57.20	81.71	89.56
VLG-RMOT (Ours)	35.68	23.75	53.71	33.20	44.68	60.54	81.41	90.28

Table 3
Quantitative results on Refer-BDD. **Bold** indicates the best result among end-to-end methods.

Method	HOTA	DetA	AssA	DetRe	DetPr	AssRe	AssPr	LocA
TransRMOT (Wu et al., 2023)	34.79	26.22	47.56	38.46	41.54	51.35	81.76	79.37
MOTRv2 (Zhang et al., 2023b)	35.35	22.88	55.24	46.35	30.33	64.19	75.47	85.30
EchoTrack (Lin et al., 2024)	38.00	28.57	51.24	36.60	54.25	55.32	85.81	86.99
LTTrack (Xu & Huang, 2026)	38.53	28.45	52.96	39.92	47.66	59.05	78.79	85.68
VLG-RMOT (Ours)	39.18	27.10	57.15	30.82	65.94	62.91	82.81	87.33

obtains a HOTA score of 52.70%, surpassing the second-best end-to-end method by 1.79%. Notably, VLG-RMOT also shows strong association accuracy, exceeding advanced two-stage approaches such as CPAny (Li et al., 2025c) and iKUN (Du et al., 2024) on AssA. Since AssA measures identity association quality, this result suggests that using vision-calibrated language to guide query initialization and refinement helps maintain expression-aware identity consistency in end-to-end tracking.

Table 2 reports the results on the Refer-KITTI-V2 (Zhang et al., 2025) dataset, which presents greater challenges with more diverse language descriptions and visual content. VLG-RMOT achieves the best HOTA, DetA, AssA, and AssRe among the compared end-to-end methods, improving HOTA over LTTrack (Xu & Huang, 2026) from 34.64% to 35.68%. Although the two-stage CPAny (Li et al., 2025c) still obtains

higher scores on several metrics, VLG-RMOT strengthens the end-to-end RMOT setting, especially in association accuracy. These results further support the effectiveness of scene-conditioned language guidance under more diverse referring expressions.

Table 3 further evaluates VLG-RMOT on Refer-BDD (Lin et al., 2024), which contains more crowded autonomous-driving scenes and a larger number of objects per video. VLG-RMOT achieves the best HOTA, AssA, DetPr, and LocA among the compared end-to-end methods, indicating improved association consistency, detection precision, and localization quality in complex driving scenes. Although EchoTrack (Lin et al., 2024) and MOTRv2 (Zhang et al., 2023b) obtain higher DetA and DetRe, respectively, our method shows a stronger overall balance between association and precise language-guided localization. These results suggest

that vision-calibrated language guidance generalizes beyond the KITTI-based benchmarks and remains useful when the scene contains dense traffic participants and diverse referring expressions.

To investigate the potential of large-scale pretrained vision-language models in RMOT tasks, we also evaluated two zero-shot approaches: Florence2 (Xiao et al., 2024) and Grounding DINO (Liu et al., 2024). To enable cross-frame tracking of identical targets, we utilized the detection outputs from these two methods as prompts for SAM2 (Ravi et al., 2025) and propagated them in video clips. The experimental results on Refer-KITTI and Refer-KITTI-V2 show that these zero-shot pipelines still lag behind methods specifically designed and trained for RMOT, with metrics comparable to early two-stage approaches. This indicates that, despite the impressive performance of pretrained vision-language models on certain visual tasks, they still exhibit limitations when handling complex tasks that require fine-grained language comprehension and multi-object visual localization.

4.4. Ablation studies

Impact of key components in VLG-RMOT. Table 4 examines how the three components contribute to the calibration-before-control pathway. The baseline without CSC, CQS, or TAG achieves 48.20% HOTA, while adding CSC alone raises HOTA to 51.09%. This 2.89-point gain shows that scene-conditioned calibration is beneficial even before the explicit query-control branches are introduced. More directly, when CQS and TAG are both retained, introducing CSC raises HOTA from 50.15% to 52.70%, indicating that the same query initialization and refinement modules are more effective with calibrated rather than raw language. With CSC fixed, adding CQS raises HOTA from 51.09% to 52.28% while improving both DetA and AssA; adding TAG further increases all three metrics, yielding 52.70% HOTA. This consistent progression shows that the two control stages contribute at different points in the query lifecycle: CQS strengthens referring-aware query initialization, and TAG uses the same calibrated signal to provide additional refinement after visual evidence has been incorporated.

Impact of the number of calibration layers in CSC. Table 5 studies how much visual calibration is needed before language is used for query control. Introducing one calibration layer raises HOTA from 50.15% to 52.12% and DetA from 38.89% to 41.56%, showing that a lightweight calibration step provides the best overall balance. Three layers increase AssA to 66.78% but reduce DetA to 39.06% and HOTA to 51.04%, while five layers further reduce HOTA to 50.08%. This trade-off indicates that deeper stacks do not yield uniformly better calibration: repeated cross-modal corrections may over-adapt the control signal to transient visual

responses, weakening detection quality despite maintaining strong association. We therefore use one layer as the compact CSC configuration that best balances detection and association.

Impact of adaptive residual scaling in CSC. To isolate the effect of residual scaling, we vary only the scaling design while keeping the one-layer CSC architecture and all other settings unchanged in Table 6 and Fig. 6. Jointly learning γ_v and γ_l achieves 52.12% HOTA and 41.56% DetA, improving HOTA by 1.97 points over the no-CSC reference, 1.40 points over the unscaled residual, and 2.98 points over the fixed-1/6 residual. Notably, the fixed-1/6 variant performs worse than the no-CSC reference (49.14% vs. 50.15% HOTA), indicating that uniform attenuation alone does not explain the benefit of CSC. The unscaled residual yields the highest AssA (66.17%) but substantially lower DetA (38.92%), resulting in lower overall HOTA than the fully learnable design. The partial-learning variants further show that learning γ_l alone outperforms learning γ_v alone by 0.92 HOTA (51.42% vs. 50.50%), while joint learning provides an additional 0.70-point gain over learning γ_l alone. These comparisons suggest that adaptive visual-to-language calibration contributes more strongly under this protocol, with language-to-visual calibration providing a complementary benefit.

The learned scaling distributions in Fig. 6(b) reveal an asymmetric calibration pattern rather than uniform scalar attenuation. Starting from 0.1667, γ_v converges to a mean of 0.0975 and a 5th–95th percentile interval of [0.0469, 0.1450]. Its 95th percentile and maximum value (0.1662) both remain below the initialization, meaning that the central 90% of values, and indeed all 256 channels, assign lower weights to the language-to-visual residual than at initialization. The distribution of γ_l is qualitatively different: its mean is 0.1340, its 5th–95th percentile interval of [0.0926, 0.1789] straddles the initialization, and its full range extends from 0.0596 to 0.2178. The visual-to-language residual is therefore attenuated overall but selectively strengthened in some channels and weakened in others relative to the initial calibration strength. Taken together, the two distributions indicate complementary modality-specific behavior: γ_v acts as a conservative regulator of the auxiliary linguistic correction to visual features, whereas γ_l provides more flexible channel-selective control over the visual correction to the language representation used by CQS and TAG. This asymmetry is consistent with the partial-learning ablation, in which learning γ_l alone outperforms learning γ_v alone by 0.92 HOTA, and further supports the importance of adaptive language-side calibration.

Impact of language feature processing in CQS. Table 7 compares four ways of using calibrated language features for query initialization. Average pooling achieves the best HOTA, DetA, and AssA, reaching 52.12%, 41.56%, and 65.43%, respectively. Its advantage is particularly evident in DetA, where the corresponding gains over all-token cross-

Table 4
Ablation of key components in VLG-RMOT.

CSC	CQS	TAG	HOTA	DetA	AssA
			48.20	36.94	62.97
✓			51.09	39.93	65.47
	✓		48.56	36.74	64.27
		✓	49.97	37.94	65.88
	✓	✓	50.15	38.89	64.75
✓	✓		52.28	40.57	67.44
✓	✓	✓	52.70	40.91	67.95

Table 5
Ablation of the number of vision-language calibration layers in CSC.

Layers	HOTA	DetA	AssA
0	50.15	38.89	64.75
1	52.12	41.56	65.43
3	51.04	39.06	66.78
5	50.08	38.22	65.71

Table 6
Ablation of channel-wise residual scaling in one-layer CSC on Refer-KITTI. “Learn.” denotes a trainable vector initialized to 1/6; numerical entries in the γ_v and γ_l columns denote fixed vectors.

Scaling design	γ_v	γ_l	HOTA	DetA	AssA
No CSC	–	–	50.15	38.89	64.75
Unscaled residual	1	1	50.72	38.92	66.17
Fixed LayerScale	1/6	1/6	49.14	38.26	63.24
Learn γ_v only	Learn.	1/6	50.50	39.30	64.99
Learn γ_l only	1/6	Learn.	51.42	40.49	65.34
Learn both	Learn.	Learn.	52.12	41.56	65.43

Table 7
Ablation of language feature processing in CQS.

Process	HOTA	DetA	AssA
Cross attn.	50.37	39.21	64.80
[CLS] token	51.02	40.76	63.94
Max pooling	50.43	39.08	65.14
Avg. pooling	52.12	41.56	65.43

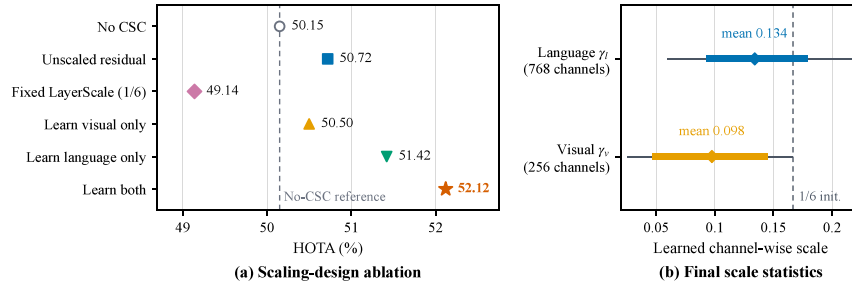


Fig. 6. Adaptive channel-wise residual scaling improves overall tracking accuracy and learns modality-specific calibration strengths. (a) HOTA of six scaling designs under the same Refer-KITTI ablation protocol. (b) Final statistics of the learned scaling vectors. Thin lines show the minimum-maximum range, thick colored lines show the 5th-95th percentile interval, diamonds denote the mean, and the dashed line denotes the shared initialization value of 1/6.

attention, the [CLS] token, and max pooling are 2.35, 0.80, and 2.48 points, respectively; it also achieves the highest AssA. The simultaneous improvements in DetA and AssA indicate that average pooling provides a more effective initialization prior for discovering referred targets without compromising association quality. The differences among the four variants help explain this result. CSC distributes scene-conditioned corrections across the language tokens, and average pooling integrates these distributed cues into a single expression-level prior. The [CLS] variant instead places the summarization burden on one token, which may underrepresent semantic cues distributed across multiple words. Max pooling independently selects the strongest response in each channel, potentially overemphasizing isolated token activations and combining extrema from different words. All-token cross-attention directly produces query-dependent language updates before the queries acquire object-specific visual evidence. Its lower performance suggests that such early interaction may prematurely entangle the common referring constraint with object-specific query differentiation. Overall, the results support aggregating all scene-calibrated tokens into a shared expression-level prior for CQS.

Impact of the position of TAG in decoder. Table 9 validates when language modulation is most effective during decoder refinement. Applying TAG before self-attention or before cross-attention achieves 50.61% and 50.70% HOTA, respectively, whereas applying it after cross-attention reaches 52.12% HOTA and the highest DetA of 41.56%. Although placement before self-attention yields a higher AssA of 66.88%, its substantially lower DetA of 38.34% limits overall tracking accuracy. The post-cross-attention placement allows TAG to evaluate queries after they have gathered visual evidence, so calibrated language can suppress expression-inconsistent responses at the stage where they are observable in the query state. This result supports post-visual modulation rather than applying the referring constraint before visual evidence has been incorporated.

Table 8

Computational cost of staged VLG-RMOT variants. All variants are profiled with the same input setting and hardware environment. HOTA on Refer-KITTI is reported for reference.

Configuration	Params (M)	FLOPs (G)	FPS	HOTA
Baseline	174.51	142.85	35.82	48.20
Baseline + CSC	180.55	156.84	33.48	51.09
Baseline + CSC + CQS	180.81	164.93	33.08	52.28
VLG-RMOT	181.21	165.05	32.90	52.70

Table 9

Ablation of the position of TAG in decoder.

Position	HOTA	DetA	AssA
Before self attn.	50.61	38.34	66.88
Before cross attn.	50.70	39.82	64.66
After cross attn.	52.12	41.56	65.43

4.5. Efficiency analysis

To assess the computational overhead of the proposed control modules, we profile staged variants under the same input setting and hardware environment. Table 8 shows that the performance gain of VLG-RMOT is obtained with a moderate computational cost. Compared with the baseline, the full model increases the number of parameters from 174.51 M to 181.21 M and the FLOPs from 142.85 G to 165.05 G, while the inference speed decreases from 35.82 FPS to 32.90 FPS. In return, HOTA improves from 48.20% to 52.70%, yielding a 4.50% absolute gain. Most of the additional parameters are introduced by CSC, which performs explicit visual calibration before language is used for query control. CQS further improves HOTA with only a small increase in parameters, and TAG brings the final improvement with negligible additional FLOPs. These results indicate that the proposed calibration-before-control design improves end-to-end RMOT accuracy while maintaining practical inference efficiency.

4.6. Visualization results

To examine whether CSC adapts language through token-specific corrections rather than uniform sentence rescaling, we visualize the relative change in the ℓ_2 norm of each token after one calibration layer. As shown in Fig. 7, object- and relation-bearing tokens such as “vehicles”, “direction”, “ours”, and “right” receive positive changes, whereas the article “the” has the largest negative change in both examples. The token “in” decreases in the first expression but increases in the second, further indicating that the adjustment is not a fixed lexical weighting rule.

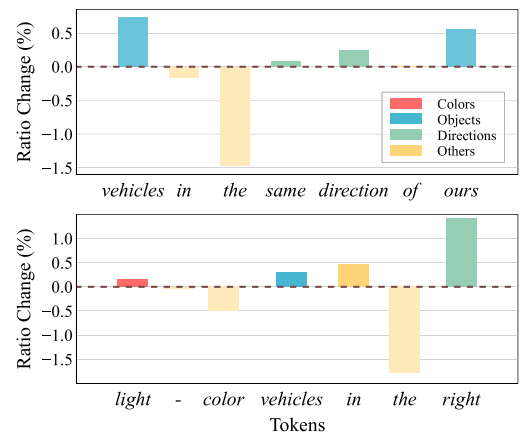


Fig. 7. Token-wise changes in language-feature norms after one CSC layer. Positive and negative values denote amplification and attenuation relative to the uncalibrated language features, respectively. The nonuniform changes illustrate how visual calibration redistributes token-level feature magnitudes across the referring expression.



Fig. 8. Qualitative comparison between TransRMOT and VLG-RMOT on Refer-KITTI. VLG-RMOT better follows fine-grained attributes and spatial constraints in the referring expression, reducing false positives among visually similar targets. Please zoom in for optimal viewing of color distinctions.

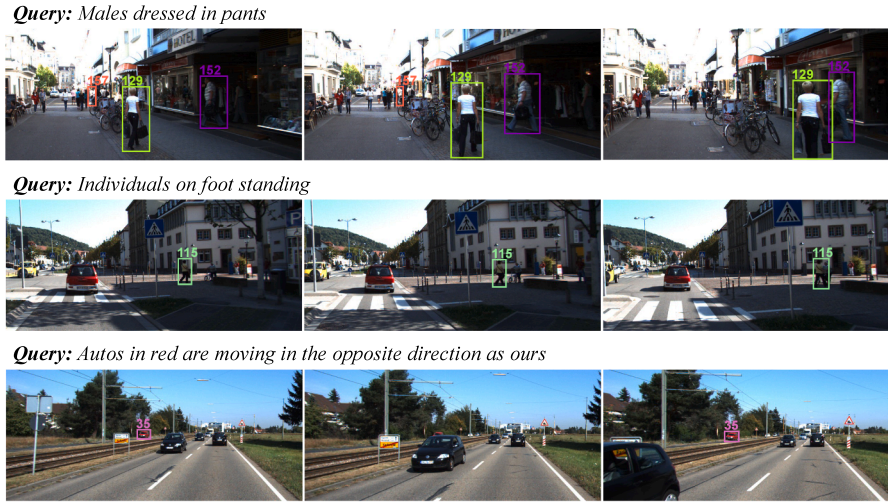


Fig. 9. Failure cases of VLG-RMOT on Refer-KITTI-V2. The examples show remaining challenges under dense similar targets, ambiguous motion or state descriptions, and severe occlusion. Please zoom in for optimal viewing.

Although feature magnitude is not a direct measure of semantic importance, these nonuniform changes provide representation-level evidence that CSC modifies the language signal according to its visual context. This evidence is consistent with the matched ablation, in which replacing raw language control with one-layer learnable CSC improves HOTA from 50.15% to 52.12% under the same protocol.

We further visualize the prediction results on Refer-KITTI in Fig. 8. TransRMOT (Wu et al., 2023) is prone to false positives when multiple targets share similar attributes (e.g., black and reddish-brown). In contrast, VLG-RMOT better follows the fine-grained attribute and spatial constraints in the referring expression. This qualitative comparison is consistent with our design: scene-conditioned language features provide more reliable query initialization, and post-visual language modulation helps filter expression-irrelevant visual responses.

5. Limitations and future work

Fig. 9 illustrates representative failure cases of VLG-RMOT. In dense crowds, the model may still confuse targets with similar clothing or body attributes, leading to missed instances or false activations. When the referring expression is semantically ambiguous, especially for motion states such as standing, walking, or moving directions, the query guidance may select visually plausible but expression-inconsistent objects. Under heavy occlusion, VLG-RMOT can temporarily lose a target when

only a very small visible region remains, even if the target is recovered after reappearing. These cases suggest that the current calibration-before-control design improves language-guided query refinement but still has limited robustness when attribute cues, motion cues, or visible evidence are weak.

Future work can address these failure modes more directly. For dense crowds, attribute-aware local calibration and hard-negative contrastive training could strengthen distinctions between visually similar people or vehicles. For ambiguous expressions, motion-state parsing or temporal relation reasoning could convert phrases such as standing, walking, and opposite direction into explicit constraints before they are used for query control. For severe occlusion, longer-term track-query memory, re-identification cues, or promptable video foundation models with streaming memory could help preserve target hypotheses through low-visibility frames. These extensions would keep the calibration-before-control principle while making the control signal more sensitive to fine-grained attributes, temporal semantics, and occlusion recovery.

6. Conclusion

VLG-RMOT introduces a calibration-before-control framework for reliable language guidance in end-to-end referring multi-object tracking. CSC adapts visual and linguistic features with learnable channel-wise residual scaling. The resulting calibrated language then guides two

complementary stages of the query lifecycle: CQS initializes the learnable queries before visual decoding, and TAG modulates the post-cross-attention query states. Across Refer-KITTI, Refer-KITTI-V2, and Refer-BDD, VLG-RMOT achieves the best HOTA among the compared end-to-end methods, while component and matched scaling ablations support the benefit of calibrated over raw language control for query initialization and refinement.

The method remains challenged by dense groups of similar targets, ambiguous motion or state descriptions, and severe occlusion, where appearance, temporal, or visible cues are insufficient. Future work will investigate finer-grained local calibration, temporal-relation modeling, and longer-term track-query memory to improve robustness in these cases.

CRedit authorship contribution statement

Hongshen Zhao: Writing – original draft, Methodology, Investigation, Conceptualization; **Ming Dai:** Writing – review & editing, Formal analysis; **Fei Xie:** Writing – review & editing, Formal analysis; **Wenkang Zhang:** Validation, Formal analysis; **Yuncong Yao:** Validation, Formal analysis; **Wankou Yang:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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